Document Title: 1.1 Brainstorming Idea
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Project: CampusAssist AI
Phase: 1. IDEATION PHASE

1.1 Brainstorming Idea

1. Project Introduction

As educational institutions become increasingly reliant on digital infrastructure, the demand for rapid, efficient, and accessible IT support for students has surged. The proposed project, **CampusAssist AI**, is a Smart IT Support Portal built on the ServiceNow platform. Its primary objective is to transform the traditional, manual IT ticketing process into an interactive, self-service experience. At the core of this solution is the **CampusCare AI** chatbot, designed to provide guided troubleshooting and context-aware escalation, bridging the gap between immediate student needs and IT support capabilities.

2. Brainstorming Context

During the ideation phase of this capstone project, the focus was placed on identifying significant inefficiencies within modern university IT infrastructures. The goal was to conceptualize a system that not only resolves technical issues rapidly but also reduces the manual overhead experienced by university IT service desks. The brainstorming sessions evaluated various technological paradigms before finalizing a ServiceNow-based architecture.

3. The Campus IT Support Problem

Through preliminary analysis, several critical pain points in existing campus IT support structures were identified:

- **High Ticket Volume for Repetitive Issues:** IT staff are overwhelmed by recurring, low-complexity requests regarding WiFi connectivity, account lockouts, and basic software installations.
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- **High Mean Time to Resolution (MTTR):** Students frequently experience prolonged wait times for trivial issues because tickets enter a generalized queue.
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- **Lack of 24/7 Interactive Support:** Existing solutions often rely on business-hour availability, leaving students without assistance during late-night study sessions.
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- **Poor Context Gathering:** Traditional forms often lack adequate detail. When students submit a ticket, IT staff must follow up to gather basic information (e.g., OS version, specific error messages), creating a frustrating back-and-forth communication loop.
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4. Initial Ideas Considered & Alternative Solutions

To address these problems, multiple conceptual solutions were proposed and evaluated:

- **Alternative A: Traditional IT Email Helpdesk**
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    - *Concept:* Consolidate all IT requests into a central email inbox using a shared ticketing tool.
    -
    - *Drawback:* Lacks self-service capabilities and does not filter or troubleshoot issues before they reach a human agent.
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- **Alternative B: Static Knowledge Base (KB) Portal**
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    - *Concept:* A website featuring a robust repository of IT articles and FAQs.
    -
    - *Drawback:* High abandonment rate; students often struggle to find the exact article for their specific context and ultimately end up calling the helpdesk anyway.
    -
- **Alternative C: Standalone Third-Party Chatbot**
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    - *Concept:* A custom web-app chatbot built on a third-party framework (e.g., Dialogflow or custom Python backend) independent of a ticketing system.
    -
    - *Drawback:* Creates a fragmented ecosystem. If the chatbot fails to resolve the issue, the context is lost, and the user must manually re-enter their information into a separate ticketing portal.
    -
- **Alternative D (Selected): Interactive Smart Portal within ServiceNow**
-
    - *Concept:* A unified Service Portal (campusassist_home) hosting a native intelligent assistant (CampusCare AI widget). It handles troubleshooting interactively and seamlessly transitions unresolved issues into native IT requests on a custom table (Student IT Request) while preserving conversation context.
    -

5. Comparison of Possible Solutions

Evaluation Criteria	**Alt A: Email Helpdesk**	**Alt B: Static KB Portal**	**Alt C: Standalone Bot**	**Alt D: CampusAssist AI (ServiceNow)**
Self-Service Capability	Low	Medium	High	**High**
Automated Troubleshooting	None	None	High	**High**

Platform Integration	Low	Low	Low	**High (Native to ServiceNow)**
Context Retention on Escalation	N/A	N/A	Low	**High (via sysparm_query)**
Development Feasibility	High	High	Medium	**High (Utilizes Glide APIs & Portal)**

6. Selected Idea

Based on the comparative analysis, **Alternative D** was selected. The project will be developed as a custom ServiceNow application scoped as **CampusAssist AI**. This approach provides an enterprise-grade platform combining frontend interactivity (Service Portal) with robust backend record management (ServiceNow tables and Glide APIs).

7. Reason for Selecting CampusAssist AI

ServiceNow was chosen as the development platform because it perfectly aligns with the requirements of an IT Service Management (ITSM) solution. By building CampusAssist AI within ServiceNow, the project benefits from:

- Pre-built UI frameworks (AngularJS-based Service Portal Designer).
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- Seamless data binding between frontend widgets and backend tables.
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- Native escalation capabilities, allowing data gathered by the chatbot to dynamically populate the custom `x_2065601_campus_0_student_it_request` table.
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- A unified ecosystem that serves both the end-user (student) and the fulfiller (IT staff).

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8. Core Features Identified During Brainstorming

To ensure the system meets the identified needs, the following core features were conceptualized for the Minimum Viable Product (MVP):

1. **CampusCare AI Chatbot:** An interactive widget (campuscare_chatbot) for immediate student engagement.
2.
3. **Categorized Troubleshooting:** Guided decision-tree logic specifically for WiFi, Hardware, Software, and Account Issues.
4.
5. **Multi-Issue Session Management:** The ability for the chatbot to maintain state during an active issue and reset once concluded, allowing a student to report multiple independent problems without cross-contamination of data.
6.
7. **Context-Aware Escalation:** Automatic passing of troubleshooting history to the support ticket if the issue remains unresolved.
8.
9. **Native Form Integration:** Redirecting the user to a ServiceNow form pre-populated using URL parameters (`sysparm_query`).
10.

9. Expected Value & Initial Project Vision

The vision for CampusAssist AI is to shift campus IT support from a reactive, manual model to a proactive, automated, and self-serving model.

- **For Students:** It offers zero-wait-time assistance for common issues and a frictionless escalation path when human intervention is necessary.
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- **For IT Staff:** It significantly deflects Level 1 (L1) support tickets, ensuring that only complex, unresolved issues reach the custom Student IT Request table, complete with pre-gathered diagnostic context.
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10. Brainstorming Conclusion

The ideation phase successfully defined a clear, impactful problem within campus IT support and evaluated multiple paradigms to solve it. By leveraging the ServiceNow platform to build CampusAssist AI and the CampusCare AI chatbot, this project will demonstrate a scalable, enterprise-ready approach to intelligent IT service delivery. The next steps will involve defining the precise problem context and mapping the user empathy landscape to formalize system requirements.

11. Brainstorming / Idea-Selection Diagram

```
[ Identify Campus IT Pain Points ]
      |
      +---> High Ticket Volume
      +---> High MTTR
      +---> Poor Context on Escalation
      |
[ Propose Alternative Solutions ]
      |
      +---> A: Email Helpdesk (Rejected: No Self-Service)
      +---> B: Static KB Portal (Rejected: Low Engagement)
      +---> C: Standalone Bot (Rejected: Fragmented Data)
      +---> D: Smart Portal in ServiceNow (Selected)
      |
[ Formulate CampusAssist AI ]
      |
      +---> Component 1: Service Portal (campusassist_home)
      +---> Component 2: CampusCare AI (campuscare_chatbot)
      +---> Component 3: Custom Table (Student IT Request)
```

End of Document Metadata

- **Document Title: 1.1 Brainstorming Idea**
- **File Name:** `1.1 Brainstorming Idea.pdf`
- **Phase: 1. IDEATION PHASE**
- **Purpose: To document the initial concept, alternative evaluations, and the strategic justification for choosing CampusAssist AI within ServiceNow for the capstone project.**
- **Recommended Diagrams: Idea Selection Process Flowchart (Provided in Section 11).**

- **Recommended Screenshots: None required for this theoretical ideation document.**